(12) INTERNATIONAL APPLICATION PUBLISHED UNDER THE PATENT COOPERATION TREATY (PCT)

(19) World Intellectual Property Organization
International Bureau

(43) International Publication Date
08 December 2022 (08.12.2022)

WIPO | PCT

(10) International Publication Number
WO 2022/254204 A1

(51) International Patent Classification:
E03C 1/04 (2006.01) *E03C 1/28* (2006.01)

(21) International Application Number:
PCT/GB2022/051387

(22) International Filing Date:
01 June 2022 (01.06.2022)

(25) Filing Language: English

(26) Publication Language: English

(30) Priority Data:
2107928.0 03 June 2021 (03.06.2021) GB

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(81) Designated States *(unless otherwise indicated, for every kind of national protection available)*: AE, AG, AL, AM, AO, AT, AU, AZ, BA, BB, BG, BH, BN, BR, BW, BY, BZ, CA, CH, CL, CN, CO, CR, CU, CZ, DE, DJ, DK, DM, DO, DZ, EC, EE, EG, ES, FI, GB, GD, GE, GH, GM, GT, HN, HR, HU, ID, IL, IN, IQ, IR, IS, IT, JM, JO, JP, KE, KG, KH, KN, KP, KR, KW, KZ, LA, LC, LK, LR, LS, LU, LY, MA, MD, ME, MG, MK, MN, MW, MX, MY, MZ, NA, NG, NI, NO, NZ, OM, PA, PE, PG, PH, PL, PT, QA, RO, RS, RU, RW, SA, SC, SD, SE, SG, SK, SL, ST, SV, SY, TH, TJ, TM,

(54) Title: DRAINAGE MONITORING APPARATUS AND SYSTEM

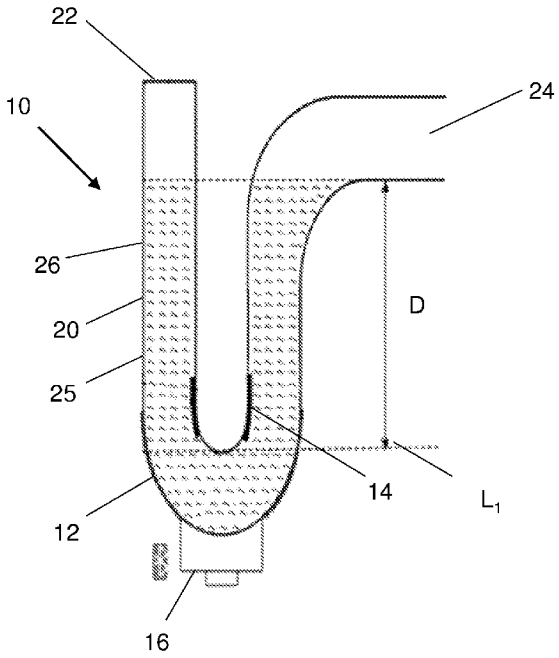


Figure 1

(57) Abstract: A drainage monitoring apparatus (10) is described comprising a water trap (20) comprising a housing (25) having a trap inlet (22) configured to be connected to a sanitary appliance outlet and a trap outlet (24) configured to be connected to waste pipe, wherein the water trap (20) is arranged to form a water seal (26) between the trap inlet (22) and the trap outlet (24); and a water level sensor (12, 14) located within the housing (25). Also described is a drainage monitoring system (100), the system (100) comprising at least one drainage monitoring apparatus (110) comprising a water trap (20) comprising a housing (25) having a trap inlet (22) configured to be connected to a sanitary appliance outlet and a trap outlet (24) configured to be connected to a waste pipe, wherein the water trap (20) is arranged to form a water seal (26) between the trap inlet (22) and the trap outlet (24), and a water level sensor (12, 14) located within the housing (25); and a water level sensor output.

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[Continued on next page]

TN, TR, TT, TZ, UA, UG, US, UZ, VC, VN, WS, ZA, ZM, ZW.

(84) **Designated States** *(unless otherwise indicated, for every kind of regional protection available)*: ARIPO (BW, GH, GM, KE, LR, LS, MW, MZ, NA, RW, SD, SL, ST, SZ, TZ, UG, ZM, ZW), Eurasian (AM, AZ, BY, KG, KZ, RU, TJ, TM), European (AL, AT, BE, BG, CH, CY, CZ, DE, DK, EE, ES, FI, FR, GB, GR, HR, HU, IE, IS, IT, LT, LU, LV, MC, MK, MT, NL, NO, PL, PT, RO, RS, SE, SI, SK, SM, TR), OAPI (BF, BJ, CF, CG, CI, CM, GA, GN, GQ, GW, KM, ML, MR, NE, SN, TD, TG).

Published:
— *with international search report (Art. 21(3))*

DRAINAGE MONITORING APPARATUS AND SYSTEM

FIELD

The present disclosure relates to a drainage monitoring apparatus and system.

BACKGROUND

Water traps are installed on every sanitary and plumbing system appliance. The water trap is installed between a fixture or appliance outlet and a waste pipe to allow for waste water drainage while forming a water seal to prevent foul odours and potentially harmful pathogens and microorganisms from entering a habitable space within the building, for example from the adjoining sewers connected to the drainage system. A water trap is generally provided as a fitting having capacity to hold a small volume of water to seal a fixture or appliance outlet from ingress of waste gases. A water trap can be provided in a variety of shapes and sizes. For example, the U-bend, S-bend, P-bend and the bottle trap. While the water level within the water trap remains at an appropriate level, the building will be sealed from the sewer system.

The loss of water from a water trap can occur in a variety of circumstances, for example, due to evaporation in an infrequently used system, or because of naturally occurring air pressure fluctuations in the building drainage system. However, the loss of the water seal can be very problematic. For example, the WHO attributed a large outbreak of the SARS virus in Amoy Garden, Hong Kong to the loss of water trap seals in a bathroom.

Despite this, the monitoring of building drainage systems and in particular, water traps is limited. Within tall or complex buildings, the drainage system may be large and complex with a number of water traps in a variety of locations. Within a conventional building drainage system, there is often little or no information relating to individual water traps and as to whether these are functioning properly. An example drainage monitoring system is disclosed in WO2008/032224 A2. This system can be used to detect defects within a building drainage system by introducing a low amplitude pressure wave into the drainage network and comparing reflected air pressure waves with a known reflected air pressure wave for that system when the system is fully functioning. However, this monitoring system can sometimes lack resolution for drainage systems with small water traps or for systems with complex branch networks.

SUMMARY

Aspects of the present disclosure relate a drainage monitoring apparatus and a drainage monitoring system.

According to a first aspect, there is provided a drainage monitoring apparatus comprising:

a water trap comprising a housing having a trap inlet configured to be connected to a sanitary appliance outlet and a trap outlet configured to be connected to a waste pipe, wherein the water trap is arranged to form a water seal between the trap inlet and the trap outlet; and

a water level sensor located within the housing.

In use, the water level sensor can be used to detect if the water level within the water trap is sufficient to form a water seal.

Beneficially, the drainage monitoring apparatus can provide real time information relating to the water level with the trap. Therefore, any defect, such as a loss of water seal can be identified and rectified. The drainage monitoring apparatus provides information relating to the water level by providing a water level sensor within the trap which operates independently from the drainage network within which the drainage monitoring apparatus is installed. Furthermore, no prior knowledge of the drainage system or the location of the water trap is required in order to obtain useful data relating to the water trap.

The water trap may be any suitable water trap. For example, the water trap may comprise a housing having a U-shape, P-shape, S-shape, bottle-shape or the like. One will appreciate that this disclosure is not limited to a particular type of water trap. The water trap may comply with required plumbing regulations. As such, the drainage monitoring system may be retrofitted into existing drainage systems by replacing the existing water traps. In this way, the drainage monitoring apparatus may be easily implemented within any building drainage system regardless of its size and complexity.

The water level sensor may be configured to continuously measure the water level in the water trap housing. The water sensor may be configured to measure the

water level in the trap housing at set intervals. For example, every 1 minute, 2 minutes, 5 minutes, 10 minutes, 30 minutes, 1 hour or any time in between.

The water level sensor may be configured to emit a signal if the water level has dropped below a pre-determined level. The water level sensor may be configured to emit a signal if the water level is less than a pre-determined level for a specified period of time, for example more than 1 minute, more than 5 minutes, more than 30 minutes. The water level sensor may be configured to emit a signal providing water level data continuously or periodically at predefined intervals. For example, every 1 minute, 2 minutes, 5 minutes, 10 minutes, 30 minutes, 1 hour or any time in between. The drainage monitoring apparatus may therefore provide an indication of the water level with a trap when there is a fault, for example the loss of a water seal or if the water level drops below a predetermined level that could be indicative of an impending loss of water seal. Additionally, the drainage monitoring apparatus may provide data relating to flow rates within the water trap, for example, frequency of use of the appliance to which the water trap is attached, volume of water usage of the appliance to which the water trap is attached.

The signal may comprise an electronic signal configured to be transmitted to a receiver. The receiver may be wired to the water level sensor. The signal may comprise an electronic signal which is configured to be transmitted to a remote receiver, for example via WiFi, Bluetooth or the like. The remote receiver may comprise, for example, a mobile phone based application, a building management system, or both. The signal may comprise an internet of things compatible signal.

The signal may be configured to generate a water level sensor output. The water level sensor output may comprise an audible output, for example, an alarm. The water level sensor output may comprise a visual output, for example, the sensor may be configured to emit a light. The water level sensor output may comprise an electronic signal configured to trigger replenishment of the water trap. For example, a remote receiver configured to receive the signal may be located on an automated valve linked to a water supply system for the water trap. The water level sensor output may comprise a data reading indicating the water level in the trap.

The water level sensor may comprise a sensing mechanism based on at least one of the following: capacitance, resistance, induction, optical, or mechanical based sensors, or any combination of these.

The drainage monitoring apparatus may comprise more than one water level sensor. A first sensor may be positioned in the housing corresponding to a lower water level and a second sensor may be positioned in the housing corresponding to an upper water level. The lower water level may correspond to a minimum water level required for a water seal. The output from the first and second sensor may comprise a difference reading between the two sensors, for example based on resistance, capacitance, induction or a combination thereof. The drainage monitoring apparatus may comprise multiple sensors positioned at multiple locations in the housing, each corresponding to a different water level.

The water level sensor or sensors may be joined to the housing of the water trap. For example, the water level sensor or sensors may be joined using a suitable adhesive. The water level sensor or sensors may be formed as part of the housing. For example, the water level sensor or sensors may be moulded into the housing of the water trap. The water level sensor or sensors may comprise discrete sensing elements. The sensing elements may be located individually or in the form of a sensing wire or strip.

The sensing mechanism may be a capacitance-based mechanism. For example, the housing may be provided with conductive plates positioned on the inside surface of the housing and the water within the housing may form a dielectric core between the conductive plates.

The sensing mechanism may be a resistance-based mechanism. For example, the housing may be provided with at least one resistive strip positioned on the inside surface of the housing and contact between the water and the resistive strip indicates a water level within the housing.

The sensing mechanism may be an induction-based mechanism. For example, the housing may comprise a transformer. The transformer may comprise a primary and secondary coil. The transformer may comprise coils wrapped around or integrated into the trap housing arranged as an electrical transformer. The housing may be provided

with at least one resistive strip positioned on the inside surface of the housing. The primary and secondary coil may be used in combination with the at least one resistive strip to provide a variable core for the transformer.

The sensing mechanism may comprise an input and an output. The sensing mechanism may rely on an applied signal to the input. The applied signal may comprise a specified frequency. For example, the applied signal may comprise an alternating signal applied at any frequency greater than zero. The applied signal may comprise an alternating signal having a frequency of preferably about 10,000 Hz. A change in the trap seal water depth may be measured based on a detected root mean square voltage (V_{RMS}) at the output when an alternating signal is applied to the input.

According to a second aspect of the disclosure, there is provided a drainage monitoring system, the system comprising:

at least one drainage monitoring apparatus comprising a water trap comprising a housing having a trap inlet configured to be connected to a sanitary appliance outlet and a trap outlet configured to be connected to a waste pipe, wherein the water trap is arranged to form a water seal between the trap inlet and the trap outlet, and a water level sensor located within the housing; and

a water level sensor output.

The drainage monitoring system may further comprise a drainage monitoring apparatus associated with every sanitary appliance within a building drainage system. The drainage monitoring system may further comprise a drainage monitoring apparatus associated with selected sanitary appliances within a building drainage system. For example, appliances that may be associated with infrequent use and are therefore more likely to be experience a loss of water seal through evaporation. Therefore, the drainage monitoring system may provide information relating to the water level within water traps throughout a building drainage system, thereby allowing for effective monitoring and rectifying of any faults within the building drainage system, for example, the loss of the water seal in a particular trap location.

The water level sensor may be configured to continuously measure the water level in the water trap housing. The water sensor may be configured to measure the

water level in the trap housing at set intervals. For example, every 1 minute, 2 minutes, 5 minutes, 10 minutes, 30 minutes, 1 hour or any time in between.

The water level sensor may configured to emit a signal if the water level has dropped below a pre-determined level. The water level sensor may be configured to emit a signal if the water level is less than a pre-determined level for a specified period of time, for example more than 1 minute, more than 5 minutes, more than 30 minutes. The water level sensor may be configured to emit a signal providing water level data continuously or periodically at predefined intervals. For example, every 1 minute, 2 minutes, 5 minutes, 10 minutes, 30 minutes, 1 hour or any time in between. The drainage monitoring apparatus may therefore provide an indication of the water level with a trap when there is a fault, for example the loss of a water seal or if the water level drops below a predetermined level that could be indicative of an impending loss of water seal. Additionally, the drainage monitoring apparatus may provide data relating to flow rates within the water trap, for example, frequency of use of the appliance to which the water trap is attached, volume of water usage of the appliance to which the water trap is attached.

The signal may comprise an electronic signal configured to be transmitted to a receiver, wherein the receiver is wired to the water level sensor. The signal may comprise an electronic signal which is configured to be transmitted to a remote receiver, for example via WiFi, Bluetooth or the like. The signal may comprise an internet of things compatible signal.

The signal may be configured to trigger the water level sensor output. Where more than one drainage monitoring apparatus is provided, the system may comprise a single water level sensor output or multiple water level sensor outputs, one output associated with each drainage monitoring apparatus. The water level sensor output may comprise an audible output, for example, an alarm. The water level sensor output may comprise a visual output, for example, the sensor may be configured to emit a light. The water level sensor output may comprise a data reading indicating the water level in the trap. The water level sensor output may comprise an electronic signal configured to activate replenishment of the water trap. For example, a remote receiver configured to receive the signal may be located on an automated valve linked to a water supply system for the water trap.

The drainage monitoring system may further comprise a receiver. The receiver may be wired to the water level sensor. The receiver may comprise, for example, a mobile phone based application, a building management system, or both.

The drainage monitoring system may further comprise at least one automated valve operatively associated with a water inlet for an appliance associated with the water trap. Where multiple drainage monitoring apparatuses are provided, the system may comprise an automated valve associated with each drainage monitoring apparatus. Alternatively, the system may comprise an automated valve associated with selected drainage monitoring apparatus within a building drainage system. For example, appliances that may be associated with infrequent use and are therefore more likely to be experience a loss of water seal through evaporation.

The automated valve may comprise a receiver configured to receive a signal from the water level sensor. The automated valve may comprise a receiver configured to receive a signal from the remote receiver (such as the mobile phone based application and/or building management system). The signal may allow for opening of the valve when the water level is below a pre-determined water level, and closing of the valve when the water level has returned to a second pre-determined level. The automated valve may be wirelessly connected to the remote receiver (such as the mobile phone based application and/or building management system), wherein a user of the remote receiver may selectively open or close the automated valve based on water level data received by the remote receiver.

For the purposes of the present disclosure, it should be understood that the features defined above or described below may be utilised, either alone or in combination with any other defined feature, in any other aspect, embodiment, or example or to form a further aspect, embodiment or example of the disclosure.

BRIEF DESCRIPTION OF DRAWINGS

These and other aspects of the present disclosure will now be described, by way of example only, with reference to the accompanying drawings in which:

Figure 1 is an example drainage monitoring apparatus having a P-shaped water trap;

Figure 2 is a graph demonstrating the effect of water level on V_{RMS} for a capacitance-based sensor;

Figure 3 is a schematic of a building drainage system with a drainage monitoring system; and

Figure 4 is a schematic of another building drainage system having a drainage monitoring system with automated valves.

DETAILED DESCRIPTION OF DRAWINGS

A drainage monitoring apparatus 10 is shown in Figure 1. A water trap 20 is provided having a trap inlet 22 configured to be connected to a sanitary appliance and a trap outlet 24 configured to be connected to a waste pipe. The water trap 20 is configured to form a water seal 26. A normally functioning water trap will have a water seal 26 with a depth D. The water seal 26 requires a minimum depth of water to form a water seal. If the water present within the trap 20 falls below L_1, there is no water seal and waste gases and air contaminated with pathogens from the adjoining sewer may be able to flow from the trap outlet 24 to the trap inlet 22 and through the adjoining sanitary appliance, to enter the habitable space. Figure 1 shows a P-shape water trap 20 although it will be appreciated that any shape of trap may be used in the present disclosure, for example U-shaped, S-shaped, bottle-shaped or any other shaped water traps.

The water trap 20 is provided with sensors 12 and 14 located on the inside wall of the water trap housing 25. In Figure 1, the sensors 12 and 14 are capacitance-based sensors provided on sensing strips or wires that have been moulded into the trap housing 25. For example, the sensors 12 and 14 may be in the form of conductive plates. However, it will be appreciated that any appropriate form of sensor may be employed provided that the sensor can monitor water level within the water trap. For example, the sensors may be capacitance-based, resistance-based, induction-based, optical or mechanical, or any combination thereof. Where a resistance-based system is desired, the sensors 12 and 14 can be in the form of resistive strips and contact with between the water and the resistive strips indicates the water level. Where an induction-based system

is desired, the water trap housing may additionally comprise coils wrapped around or integrated into the trap housing arranged as an electrical transformer. For example, primary and secondary coils can be formed within the housing and which are configured to be used in combination with sensors 12 and 14 in the form of resistive strips.

Figure 2 demonstrates the effect of changing trap seal water depth on the detected root mean square voltage (V_{RMS}) at the output when an alternating signal is applied to the input. The applied signal can be of any frequency greater than zero, optimally 10,000 Hz. . The sensors 12 and 14 are configured to operate continuously and emit readings via electrical circuit 16 which is connected via Bluetooth to a remote receiver. It will be appreciated that any form of wireless communication may be employed, for example via a building WiFi network. The remote receiver may comprise a mobile phone application or a central building management system. Sustained low voltage readings, for example, over the course of several minutes may indicate that the water seal 26 within the water trap 20 has been lost and the user can be alerted via an alert via the mobile phone application or the central building management system. Alternatively, electrical circuit 16 may be provided with a direct output, such as an alarm or a light which will be triggered by sustained low voltage readings. The alarm or light may remain active until the water within the water trap 20 has been replenished to form a water seal 26.

Where the drainage monitoring apparatus 10 is configured for use with a remote receiver, the readings from sensors 12 and 14 may be transmitted continuously allowing a user to accumulate data relating to the water trap usage. This may provide for improved water management within a building. The drainage monitoring apparatus 10 may be internet of things ready, facilitating ease of use with complementary systems, such as building management systems.

A schematic building drainage system 100 is shown in Figure 3. Within the building drainage system 100, each sanitary appliance 140 is provided with a drainage monitoring apparatus 110 according to the present disclosure. The drainage monitoring apparatus 110 is connected between the appliance 140 and a branch waste pipe 134, which is in turn connected to the main waste pipe 130 leading to sewers 132.

Each drainage monitoring apparatus 110 within the drainage system is provided with a water trap having at least one water level sensor configured to emit a signal via Bluetooth to a remote receiver 150. The signal may be emitted periodically, continuously, or when the water level within the trap falls below a minimum water level required to form a water seal within the trap. The remote receiver 150 may comprise, for example, a mobile phone based application or a building management system. The building management system may comprise, for example, a central alarm box or a central computer.

Alternatively, or additionally, each sensor within each drainage monitoring apparatus 110 can be configured to continuously emit water level data. This data can be stored and analysed on the remote receiver allowing a user to obtain accurate information relating to the water usage of each appliance within the building drainage network 100. The remote receiver 150 can be configured to emit an alert when readings indicate that the water seal within one or more of the water traps within the building have lost their water seal.

A further drainage system 200 is shown in Figure 4. This system is similar to that shown in Figure 3 and where appropriate like reference numerals, incremented by 100 have been used. Within the building drainage system 200, each sanitary appliance 240 is provided with a drainage monitoring apparatus 110 according to the present disclosure. The drainage monitoring apparatus 210 is connected between the appliance 240 and a branch waste pipe 234, which is in turn connected to the main waste pipe 230 leading to sewers 232.

Each drainage monitoring apparatus 210 within the drainage system is provided with a water trap having at least one water level sensor. Additionally, each drainage monitoring apparatus 210 is provided with an automated valve 242 which is connected to a water supply for the appliance 240. The water level sensor within each water trap is configured to open the automated valve 242 when the level within the water trap falls below a minimum level required for a water seal. In this example, the drainage monitoring apparatus controls the automated valve via Bluetooth. It will be appreciated that the drainage monitoring apparatus could be connected directly to the valve (for example, via cabling) or could communicate over another network such as a building WiFi network. The drainage monitoring apparatus 250 is configured to close the automated valve 242

when the water level has returned to a predetermined level, for example the maximum water seal depth (D). The drainage monitoring apparatus 210 can also emit an alert to a remote receiver 250 indicating at least one of the following: that a valve 242 within the system 200 has been activated; that the water level with a trap has fallen below a minimum water level required to form a water seal within the trap; continuously emit water level readings from each drainage monitoring apparatus 210. This data from each drainage monitoring apparatus 210 can be stored and analysed on the remote receiver 250 allowing a user to obtain accurate information relating to the frequency of use and the water usage of each appliance within the building drainage network 200. The remote receiver 250 can be configured to emit an alert when readings indicate that the water seal within one or more of the water traps within the building have lost their water seal. Alternatively or in addition, the automated valve 242 may be operated via the remote receiver. For example, a user may select to open or close a valve 242 associated with an appliance based on water level readings received from drainage monitoring apparatus.

It should be understood that the examples provided herein are merely exemplary of the present disclosure and that various modifications may be made thereto without departing from the scope defined by the claims.

CLAIMS

1. A drainage monitoring apparatus comprising:

a water trap comprising a housing having a trap inlet configured to be connected to a sanitary appliance outlet and a trap outlet configured to be connected to waste pipe, wherein the water trap is arranged to form a water seal between the trap inlet and the trap outlet; and

a water level sensor located within the housing.

2. The drainage monitoring apparatus of claim 1, wherein the water level sensor is configured to continuously measure the water level in the water trap housing.

3. The drainage monitoring apparatus of claim 1, wherein the water sensor is configured to measure the water level in the trap housing at set intervals.

4. The drainage monitoring apparatus of any of claims 1 to 3, wherein the water level sensor is configured to emit a signal if the water level within the trap drops below a pre-determined level.

5. The drainage monitoring apparatus of any of claims 1 to 4, wherein the water level sensor is configured to emit a signal providing water level data continuously or periodically at predefined intervals.

6. The drainage monitoring apparatus of any of claims 4 or 5, wherein the signal is an electronic signal transmitted to a receiver, wherein the receiver is wired to the water level sensor.

7. The drainage monitoring apparatus of any of claims 4 or 5, wherein the signal is an electronic signal configured to be transmitted to a remote receiver.

8. The drainage monitoring apparatus of any of claims 4 to 7, wherein the signal is configured to generate a water level sensor output.

9. The drainage monitoring apparatus of claim 8, wherein the water level sensor output comprises an audible output.

10. The drainage monitoring apparatus of claim 8 or 9, wherein the water level sensor output comprises a visual output.

11. The drainage monitoring apparatus of any of claims 8, 9 or 10, wherein the water level sensor output comprises an electronic signal configured to trigger replenishment of the water trap.

12. The drainage monitoring apparatus of claim 11, further comprising an automated valve configured to be connected to a water supply for an appliance associated with the water trap.

13. The drainage monitoring apparatus of any preceding claim, wherein the water level sensor comprises a sensing mechanism based on at least one of the following: capacitance, resistance or induction.

14. The drainage monitoring apparatus of any preceding claim, comprising more than one water level sensor.

15. The drainage monitoring apparatus of any preceding claim, comprising a first water level sensor positioned in the housing corresponding to a lower water level and a second sensor positioned in the housing corresponding to an upper water level.

16. The drainage monitoring apparatus of any preceding claim comprising multiple sensors positioned at multiple locations in the housing, each location corresponding to a different water level.

17. A drainage monitoring system, the system comprising:

at least one drainage monitoring apparatus comprising a water trap comprising a housing having a trap inlet configured to be connected to a sanitary appliance outlet and a trap outlet configured to be connected to a waste pipe, wherein the water trap is arranged to form a water seal between the trap inlet and the trap outlet, and a water level sensor located within the housing; and

a water level sensor output.

18. The drainage monitoring system of claim 17, further comprising a drainage monitoring apparatus associated with selected sanitary appliances within a building drainage system.

19. The drainage monitoring system of any of claims 17 to 18, further comprising a receiver configured to receive the water level sensor output.

20. The drainage monitoring system of any of claims 17 to 19, further comprising at least one automated valve configured to be connected to a water inlet for an appliance associated at least one water trap.

21. The drainage monitoring system of claim 20, wherein the automated valve comprises a receiver configured to receive a signal from the water level sensor that opens and closes the valve according to the water level within the trap.

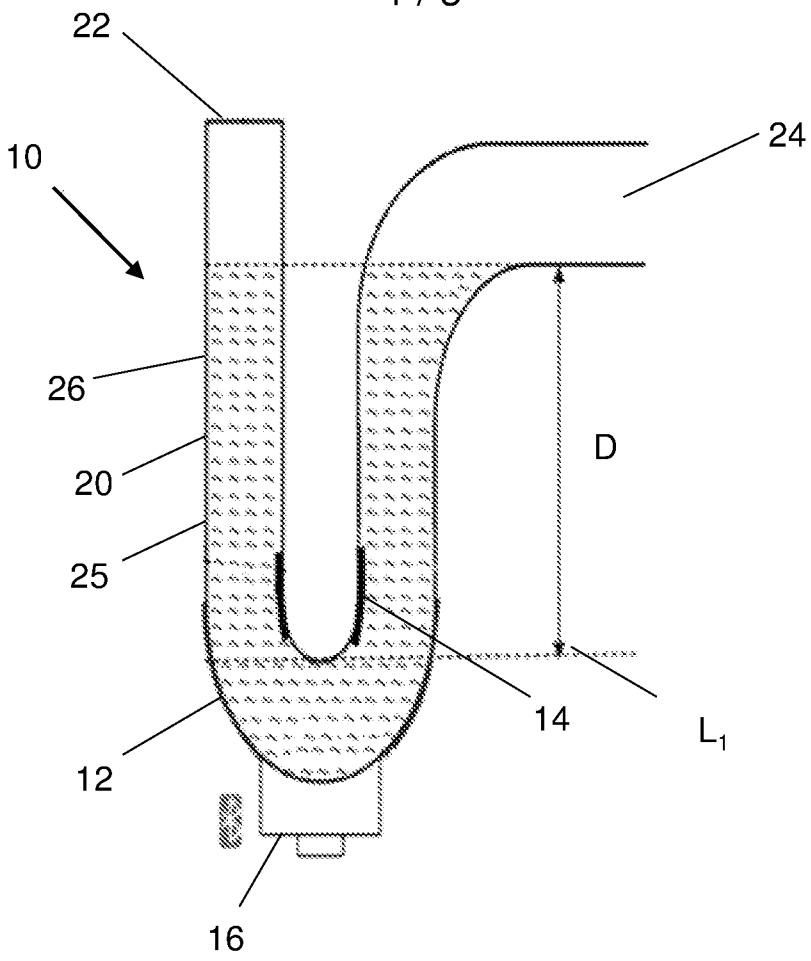


Figure 1

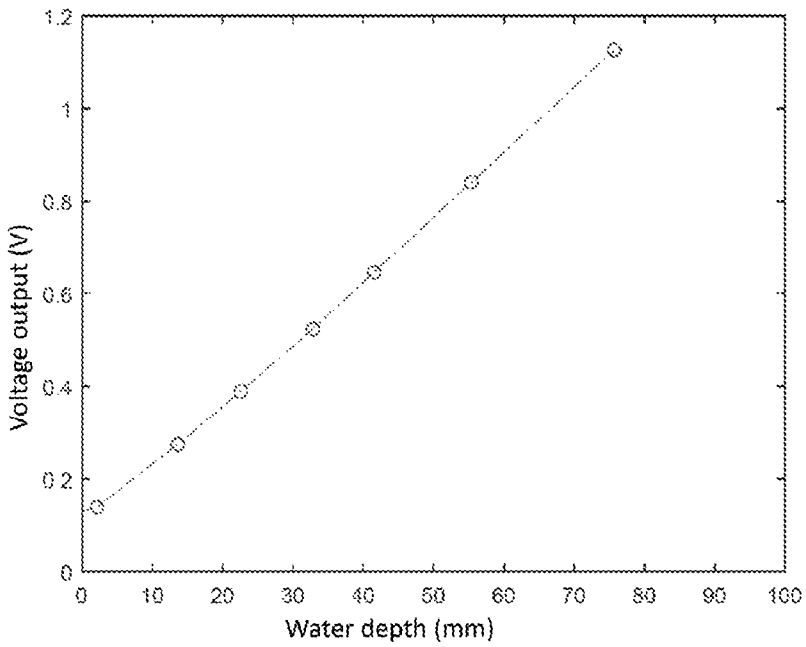


Figure 2

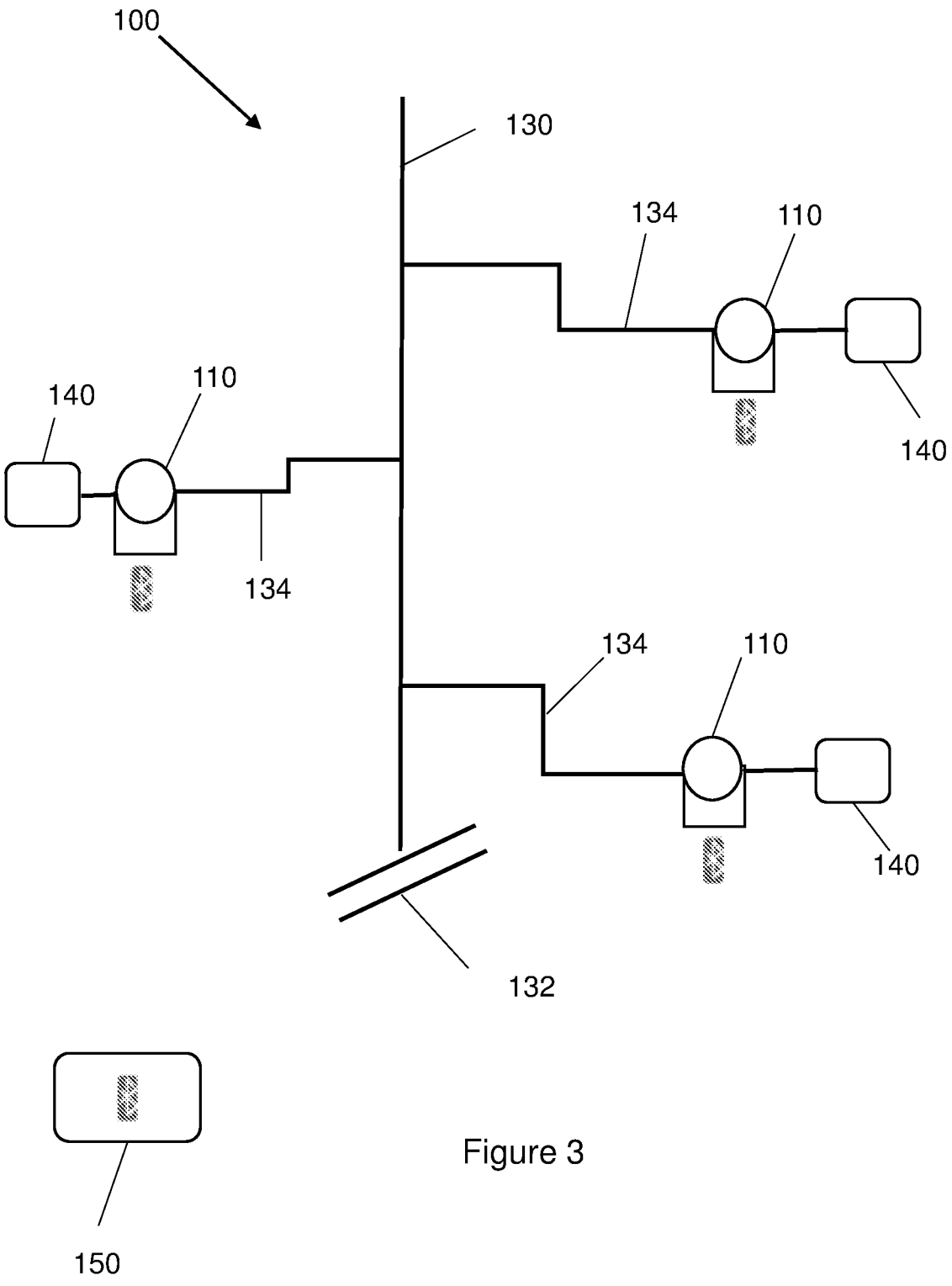


Figure 3

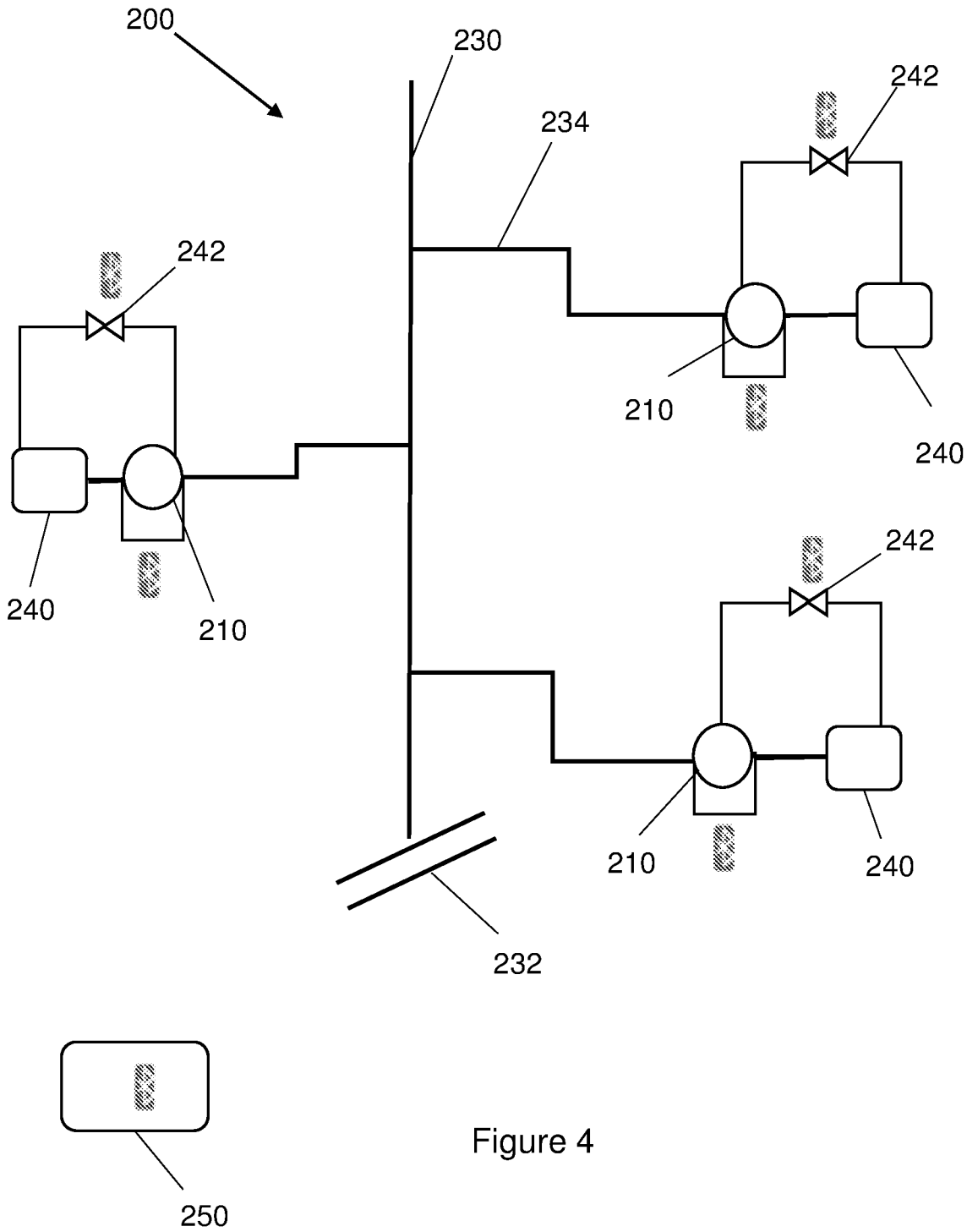


Figure 4

INTERNATIONAL SEARCH REPORT

International application No
PCT/GB2022/051387

A. CLASSIFICATION OF SUBJECT MATTER
INV. E03C1/04 E03C1/28
ADD.

According to International Patent Classification (IPC) or to both national classification and IPC

B. FIELDS SEARCHED

Minimum documentation searched (classification system followed by classification symbols)
E03C

Documentation searched other than minimum documentation to the extent that such documents are included in the fields searched

Electronic data base consulted during the international search (name of data base and, where practicable, search terms used)
EPO-Internal

C. DOCUMENTS CONSIDERED TO BE RELEVANT

Category*	Citation of document, with indication, where appropriate, of the relevant passages	Relevant to claim No.
X	CN 108 978 811 A (SHENZHEN SPRING WOODS HOLDINGS CO LTD) 11 December 2018 (2018-12-11)	1,3,9-21
A	figures 1, 2	2,4-8
X	DE 90 04 405 U1 (N.N.) 28 June 1990 (1990-06-28) page 2, paragraph 1; figure 1	1-12, 17-21
X	EP 1 762 791 B1 (VAILLANT GMBH [DE]) 1 April 2009 (2009-04-01) paragraph [0018]; figures 2-4	1,3,9, 10,17-21
A	US 2010/192295 A1 (FIMA GIOVANNI [US]) 5 August 2010 (2010-08-05) paragraphs [0066], [0068]; figure 1A	1-21

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[x] Further documents are listed in the continuation of Box C.

[x] See patent family annex.

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Date of the actual completion of the international search	Date of mailing of the international search report
27 July 2022	05/08/2022

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International application No
PCT/GB2022/051387

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